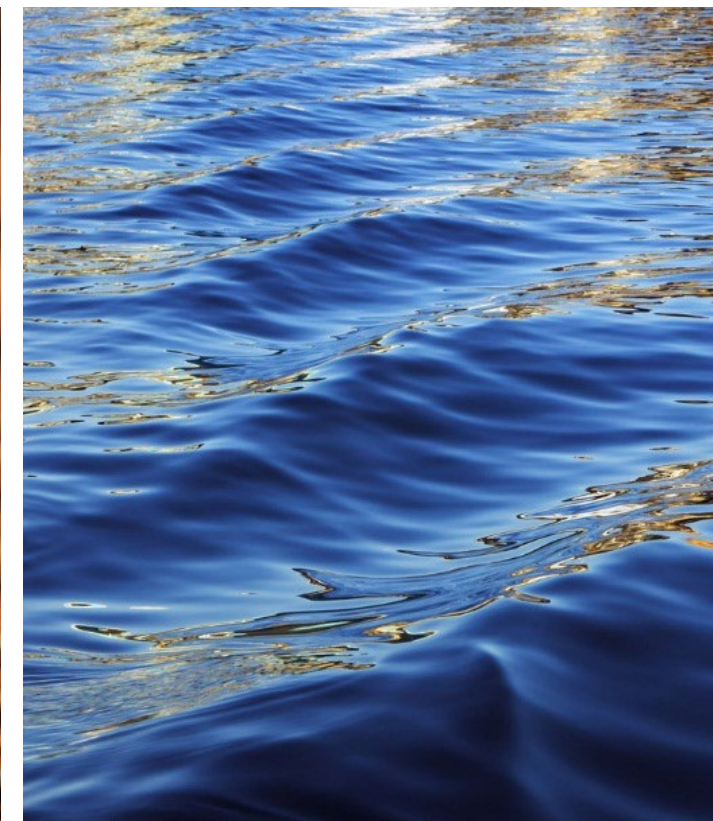
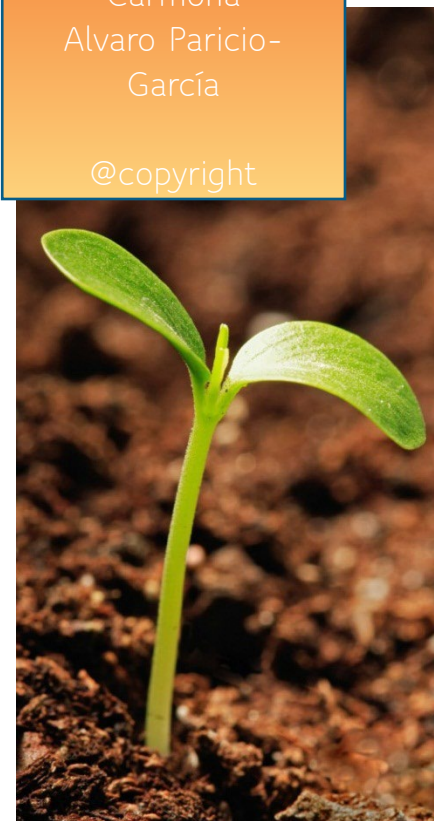




Intro to AnyLogistix (ALX)



EUGLOH

European University Alliance for Global Health



Co-funded by the
European Union



Universidad
de Alcalá



Learning objectives

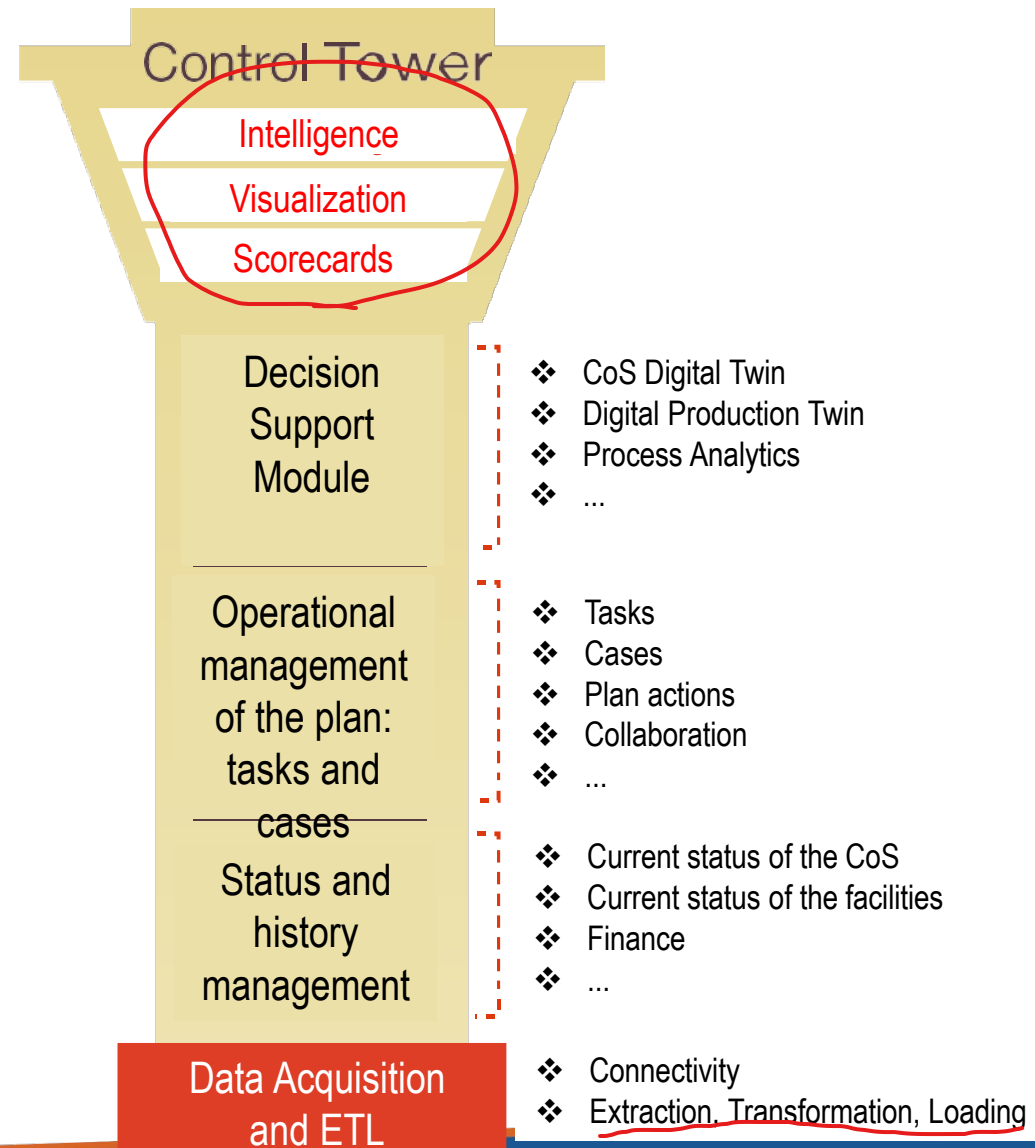
Learning objectives

- Install the AnyLogistics tool.
- Learning how to use the tool
- Work plan

Reminder: the Control Tower of the SCM

The set of systems required to provide end-to-end visibility of the entire Supply Chain.

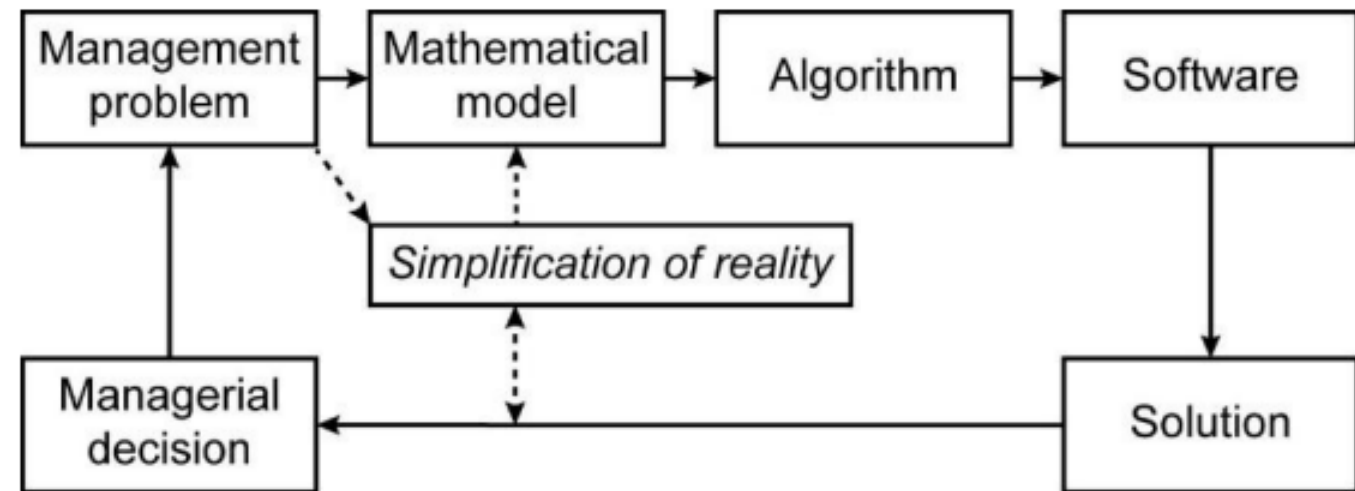
- ☐ Structured by levels.
- ☐ Holistic approach: can be extended to the entire organization.
- ☐ Includes Digital Twin:
 - Several D. twins can cohabit simultaneously.



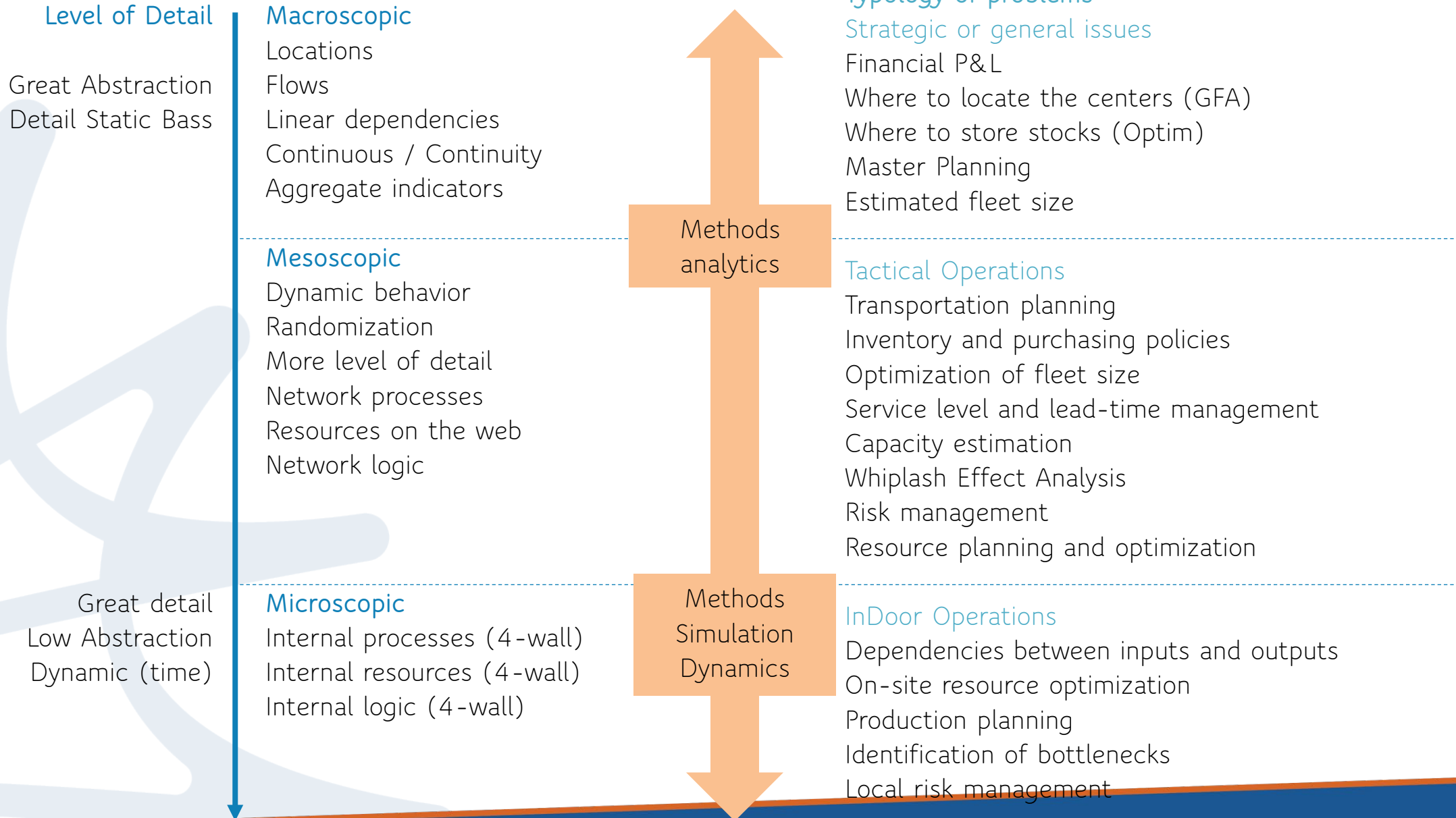


The objective of a simulation

- Model a supply chain problem to find its possible solutions and alternatives, with a view to making a real implementation decision or a mere "what-if" study.
- The starting data can be:
 - Actual data
 - Data from a study
 - Future projection
 - Alternative scenarios

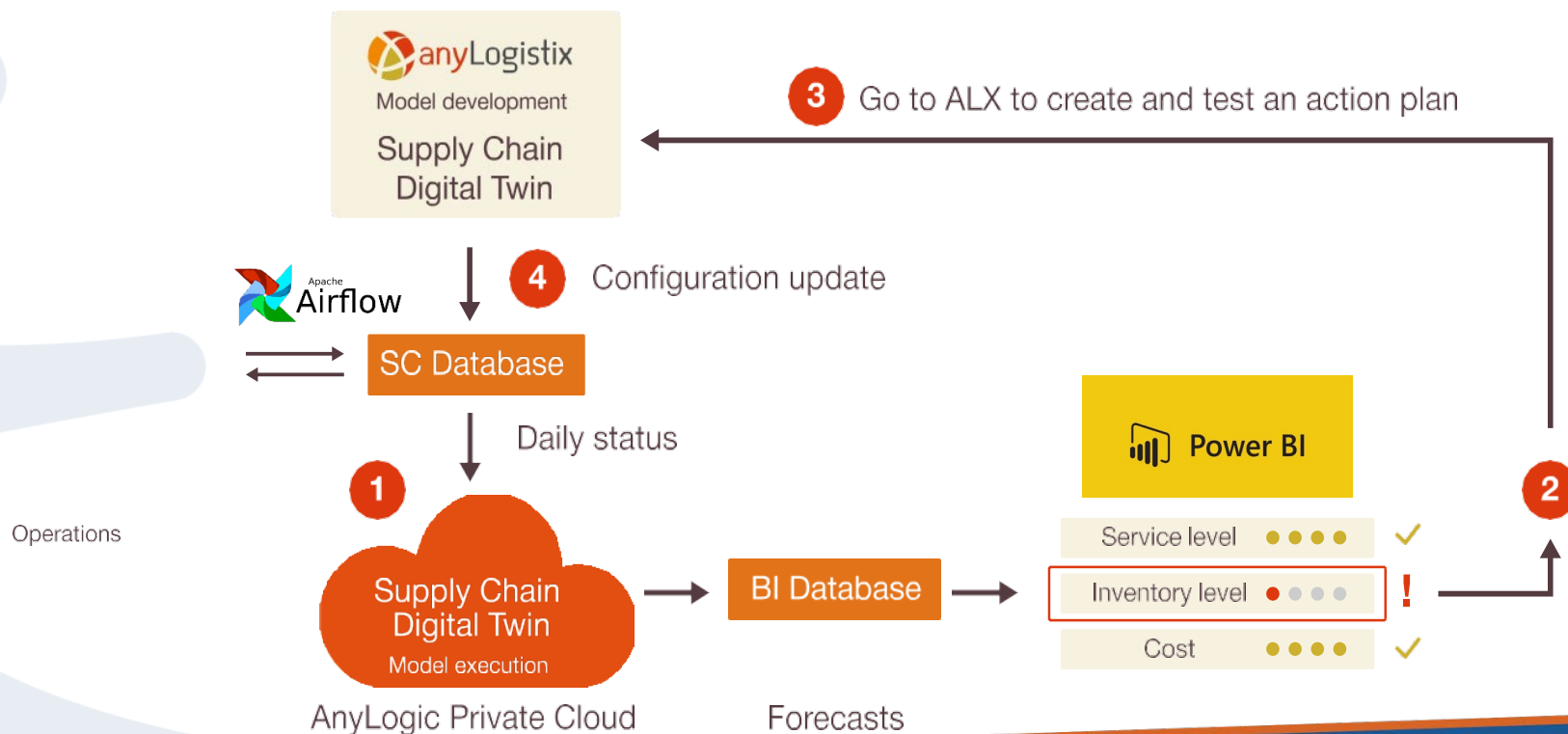


Analytical methods versus simulations





We also recall our target technology model of Digital Twin

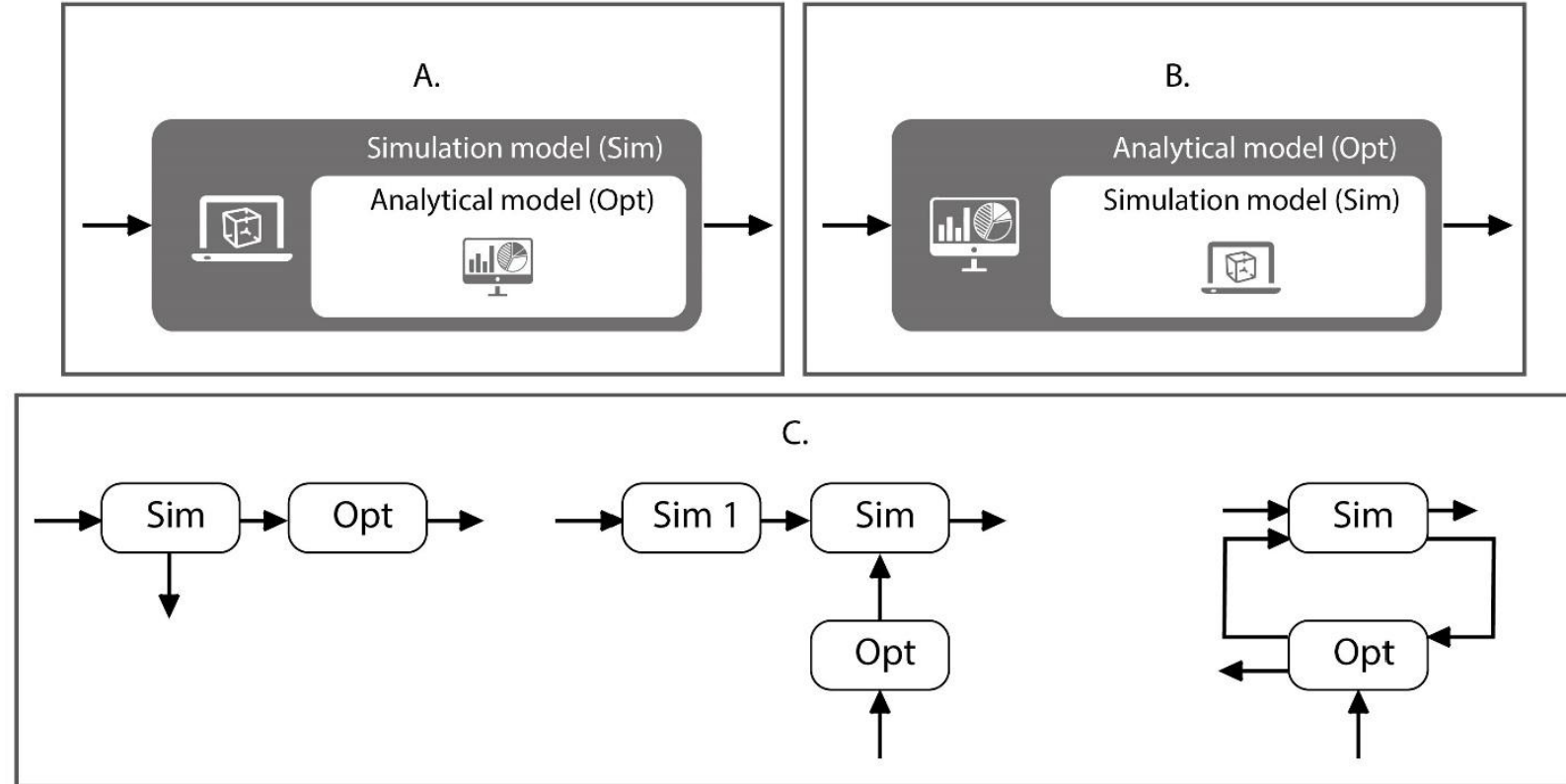




Working models with ALX

Simulation allows several approaches:

- A. Optimization as a starting point and simulation as an extended analysis method e.g. to validate analytically obtained solution
- B. Simulation as a starting point and optimization as an extended analysis method e.g. to find optimal parameter values in a simulation model
- C. Hybrid simulation-optimization approach for the improvement of supply chain





- SCM modeling and simulation tool.
- Two versions:
 - Free, stand-alone.
 - Commercial, with cloud support and con



Subscription \$20,000/year	Perpetual license \$55,000
The price does not include VAT, sales tax, withholding tax or any other tax(es) or fee(s) which may be applicable in the country of purchase.	The price does not include VAT, sales tax, withholding tax or any other tax(es) or fee(s) which may be applicable in the country of purchase.
FEATURES • Full Professional Edition • Regular updates • Advanced technical <u>support</u> • Multiple year/user discount	FEATURES • Full Professional Edition • First year of updates and advanced technical <u>support</u> included • Subsequent advanced <u>support</u> and updates \$11,000/year • Multiple user discount



Personal
Learning Edition
(PLE)

Commercial
Edition (CE)

Personal
Learning Edition
(PLE)

Commercial
Edition (CE)

EXPERIMENTS

Greenfield Analysis

- Multi-echelon
- Current network expansion (Brownfield analysis)
- GFA with roads

✓

✓

✓

✓

✓

Network Optimization

- Supply chain design
- Master planning
- Shipment scheduling

✓

✓

✓

✓

✓

✓

Simulation

- What-if analysis
- Budgeting
- Risk analysis
- Safety stock estimation
- Facility operations modeling

✓

✓

✓

✓

✓

✓

✓

✓

OTHER

Built-in IBM ILOG CPLEX® analytical solver

✓

✓

Built-in AnyLogic® simulation engine

✓

✓

Database types supported

internal database

PostgreSQL

GIS maps and database

✓

✓

Unlimited model size

✓

Multi-user access

for Linux Server

Open API (Java, Python)

for Linux Server

Advanced modeling capabilities

✓

- Custom constraints
- Demand forecast
- Fleet size setting
- Indicator constraints
- Linear ranges
- Objective members
- Path selection mode
- MRP (Material Requirements Planning) inventory policy
- Custom policies



Download AnyLogistix

The first thing is to check that it meets the necessary requirements, and the second is to download it.

<https://www.anylogistix.com/downloads/>

System requirements

anyLogistix is a Java based application and has been tested on the following platforms:

- Microsoft Windows 10, x64
- Apple macOS 13, x64

Hardware recommendations:

- anyLogistix requires 2GB of free disk space.
 - We recommend having 16GB (or more) of memory and modern processor for optimal performance.
 - Java 2 Standard Edition 10.0 or later is needed to run anyLogistix model development environment.
 - An Internet connection is required.
- Java Runtime Environment is included in the anyLogistix installation package.

DOWNLOAD ANYLOGISTIX

Latest version: **3.1.1**
Released on: **February 13, 2024**

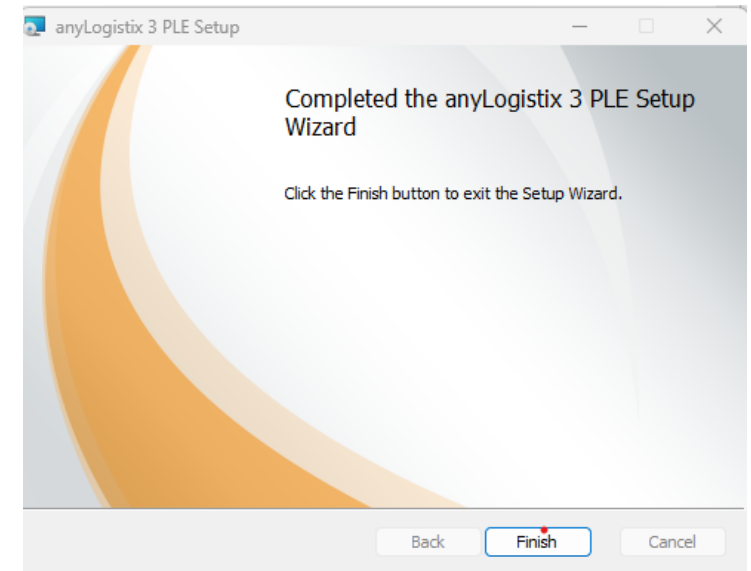
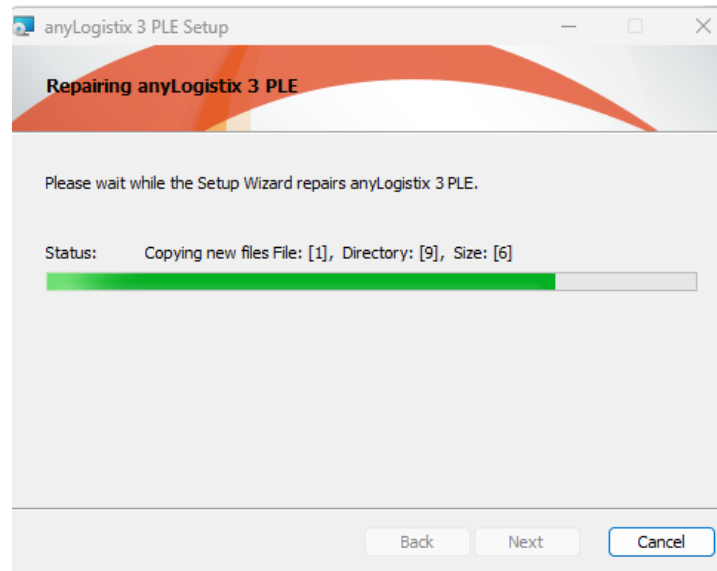
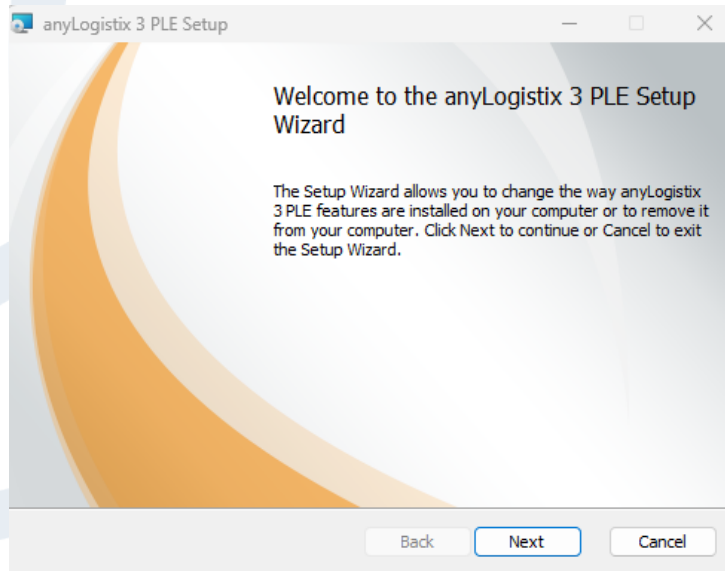
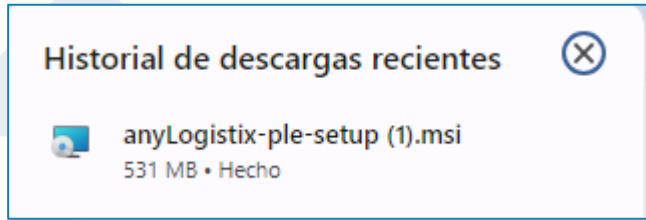
[Technical datasheet](#) | [System requirements](#) | [License agreement](#) | [Version history](#)

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AnyLogistix Installation

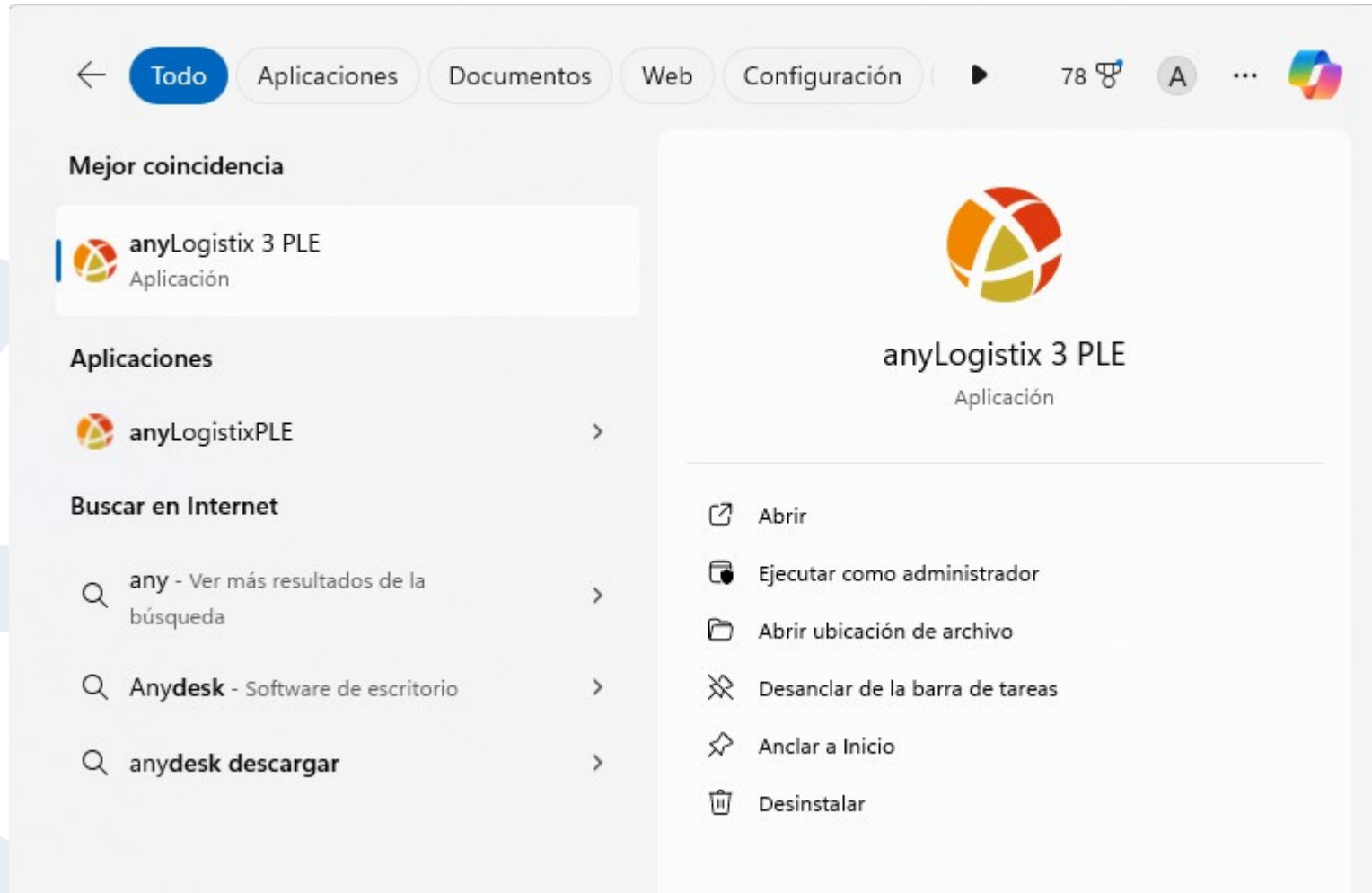
- After registering as a user, we download the "Personal Edition".





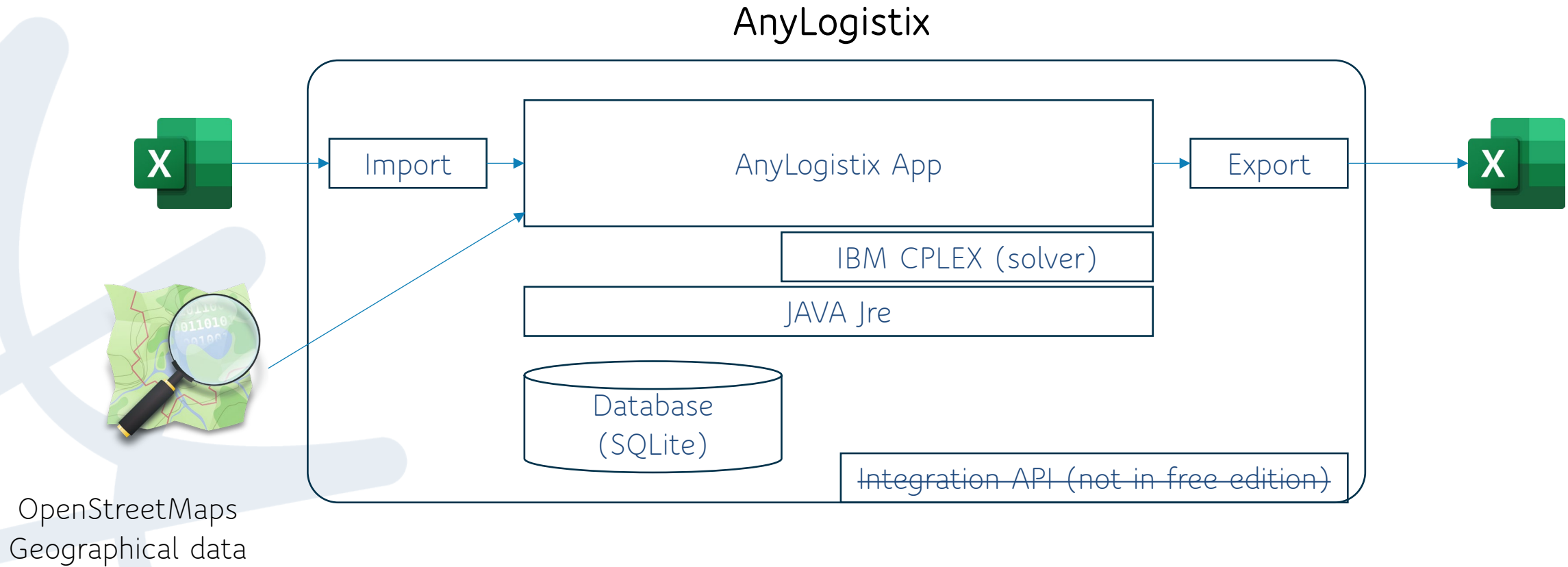
Getting started with AnyLogistix

- We open the application...





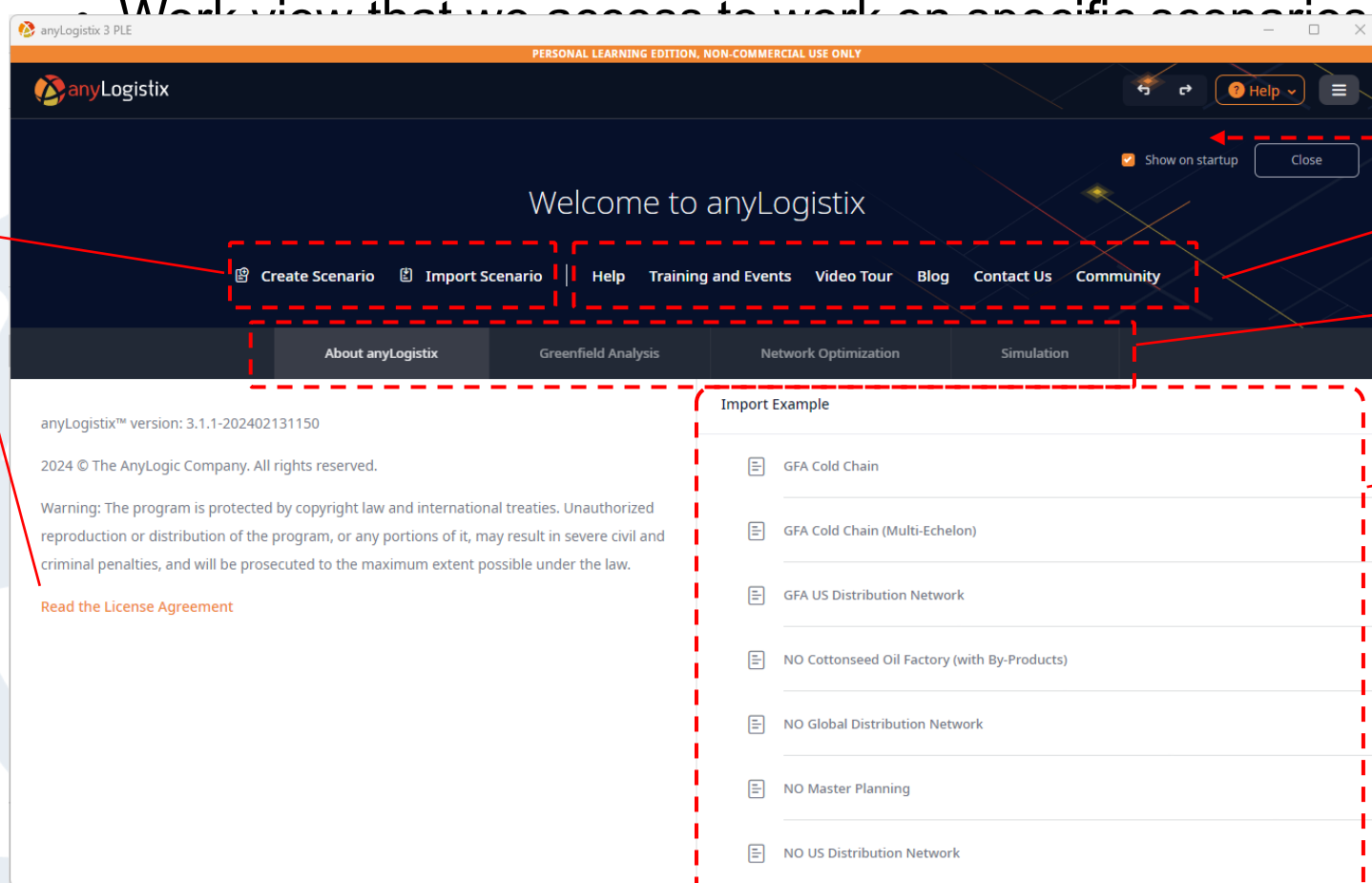
AnyLogistix (PLE) inside...





Getting started with AnyLogistix

- There are two "main screens":
 - An introductory, commercial one (below) that opens the first time. We can deactivate it...



Start working on a new scenario

Commercial information and service agreement

Access to training information, blogs and more.

Thematic filtering on predefined examples, tutorials and blogs.

Predefined examples



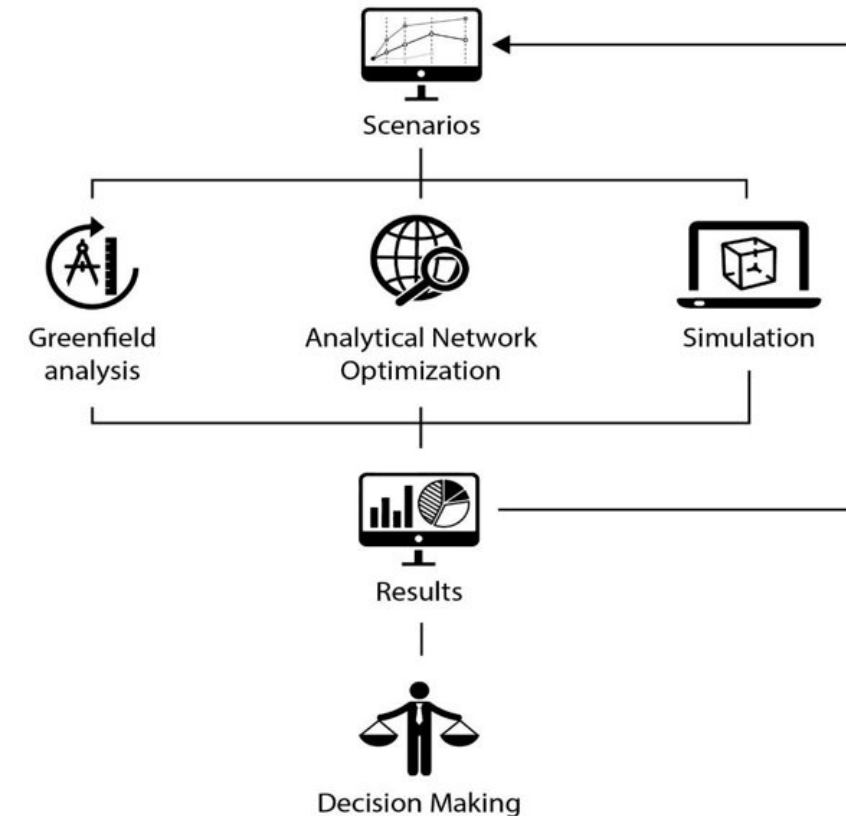
Getting started with AnyLogistix

- The tool allows working on three types of typical supply chain issues (scenarios):

1. GFA (Green-Field) Analysis

2. Supply Network Optimization

3. Simulation with parameter variation





Types of problems it facilitates

- Where are the best locations for our warehouses, distribution centers and production sites?
- How to allocate demand and supply in a complex network?
- What are the best replenishment, sourcing and transportation policies?
- How resilient is our supply chain to supplier disruptions?
- What will happen if we change our inventory policy?
- What will happen if we change the capacity of a distribution center?
- What will happen if demand changes?
- What will happen if we add a new product?
- How much does a stock out cost?



1. GFA (Green-Field) Analysis

- Solving the problem of where to locate our production and distribution centers.
- The information it uses is:
 - Customer locations/sites
 - List of products
 - Aggregate demand for each customer and product
 - Direct distance between customers and distribution centers/warehouses
 - Number of installations we need to position.



2. Supply chain optimization

- Allows:
 - Locating optimal locations for distribution and production facilities
 - Model product flows
 - Manage different sourcing options
- How to identify the optimal solution?
 - The one that offers the best set of flows and facilities for
 - ❖ Maximizing profits
 - ❖ Complying with all restrictions, especially production restrictions
- The information it uses is:
 - Demand.
 - Location of suppliers, customers, existing and potential facilities, and routes between supply chain elements.
 - Product flows.
 - Product warehousing.
 - SCM fixed and variable costs and transportation costs.
 - Time periods



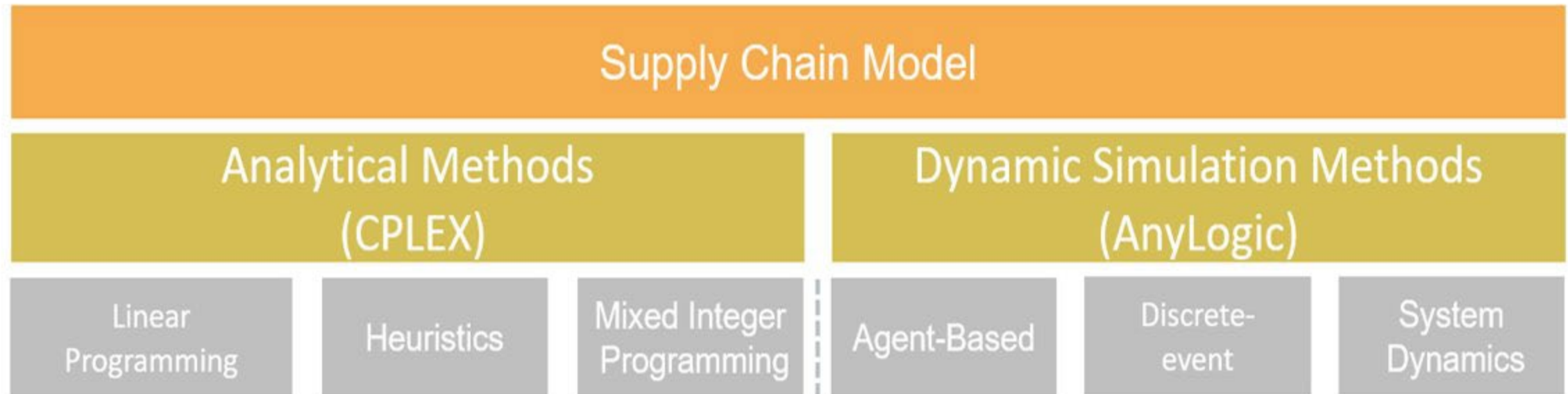
3. Simulation and Variation Experiments

- Model the actual delivery of products on the GIS map.
- Extraction of detailed statistics with temporal baseline.
- Make variation of hypotheses to compare results under different scenarios.
- It employs the same data set as GFA and network optimization, adding:
 - Suppliers
 - Product Procurement
 - Inventory policies
 - Expenses incurred



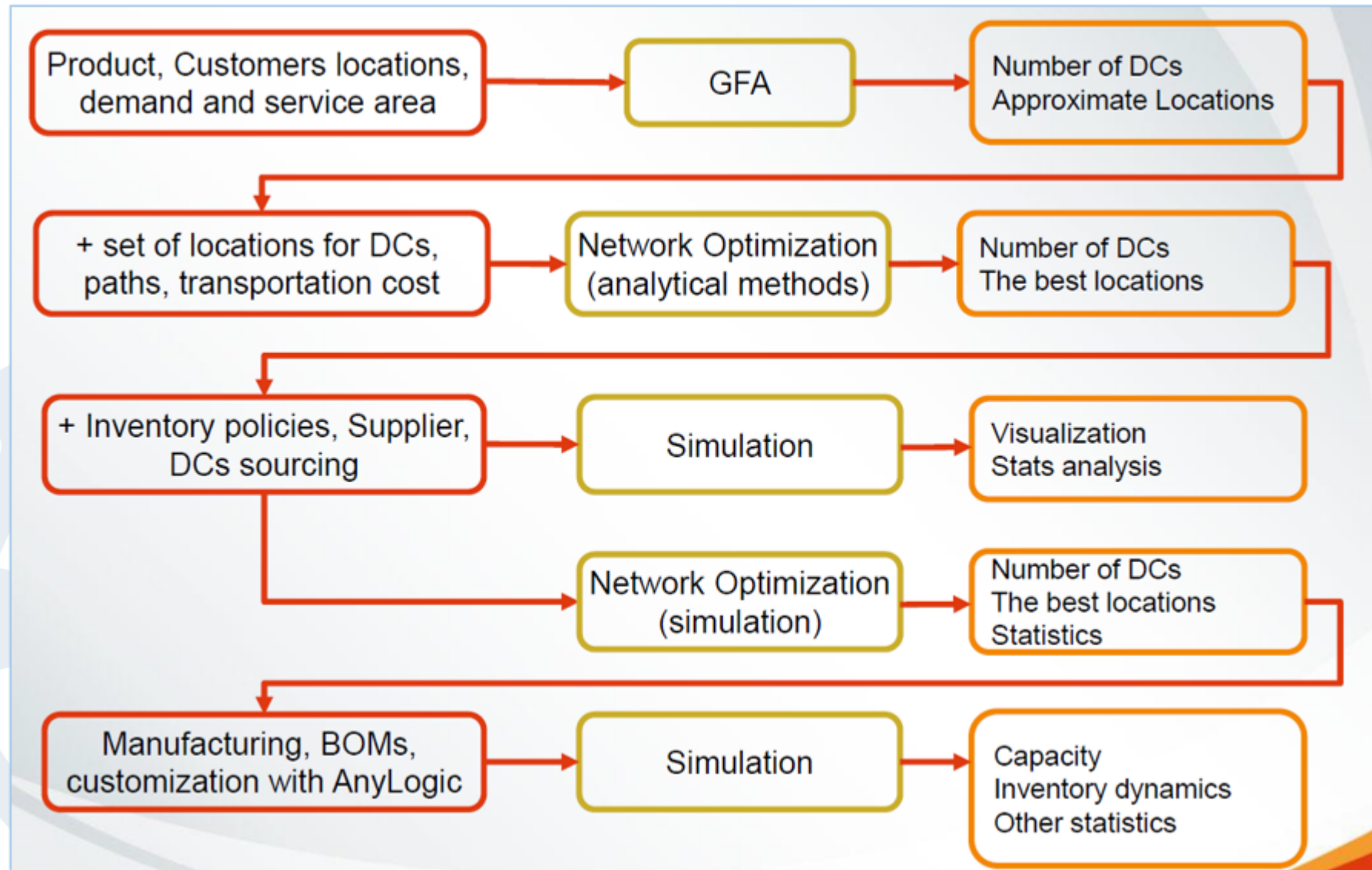
Simulation and Optimization Models

anyLogistix





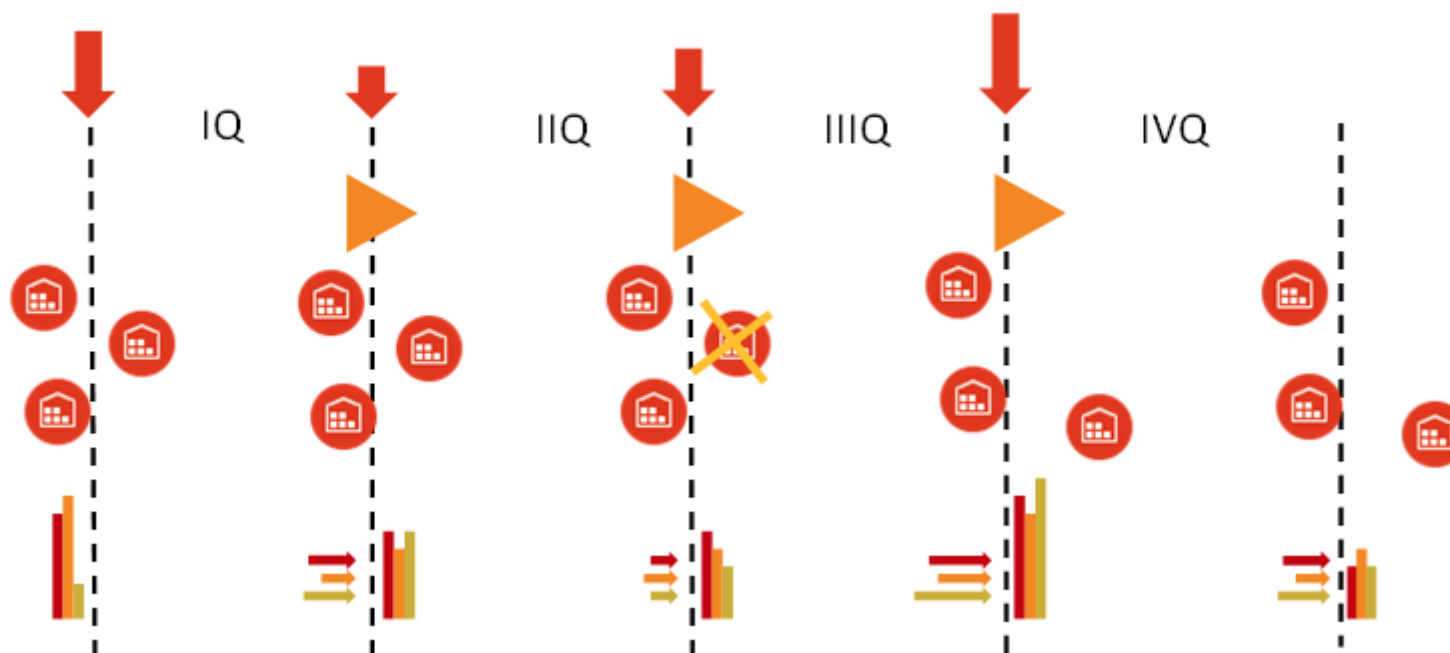
The usual ALX workflow





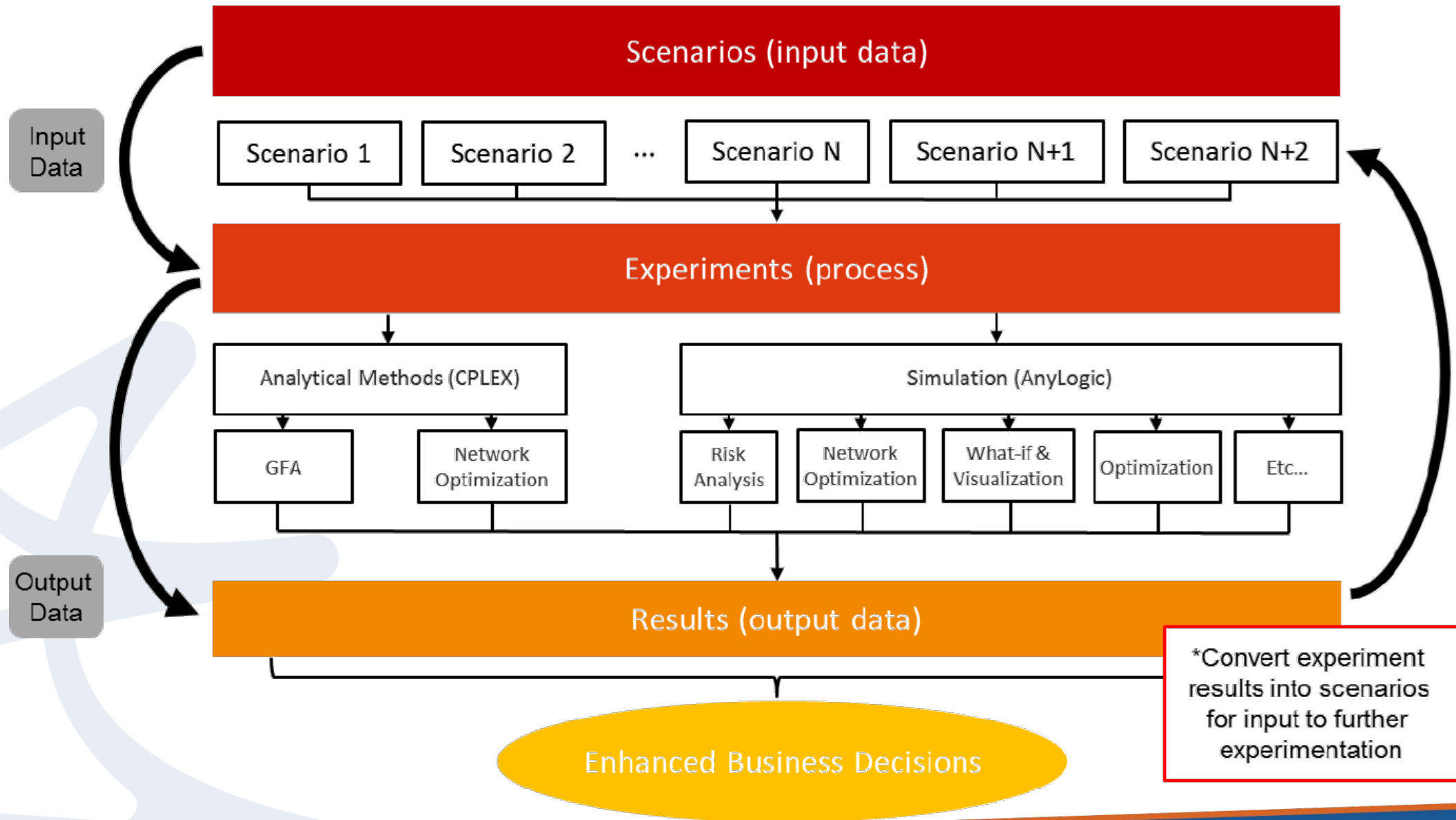
Discrete Optimization and Simulation

- Discretization:
 - Time periods are defined: day, week, month, quartile, year, ...
 - The demand is defined independently for each period.
 - The output of one period is the input of the next.
 - Inventory is planned for the beginning and end of each period.





How will we operate with ALX?





General view of the work area

Types of experiments performed

Open projects

The screenshot shows the anyLogistix 3 PLE interface. At the top, there's a header with the anyLogistix logo and a navigation bar with tabs for 'Greenfield Analysis (3)', 'Network Optimization (1)', and 'Simulation (2)'. Below the navigation bar, there are tabs for 'Cold Chain', 'Cold Chain 3', and 'Caso 1b'. The main area features a map of the UK with numerous blue location icons. On the left, a sidebar contains 'GENERAL DATA' with a 'Cold Chain' section and 'EXPERIMENTS' with 'GFA experiment 2' and 'GFA with roads experiment'.

Experiments carried out on the project

Zoomable map

Entities available for a certain experiment

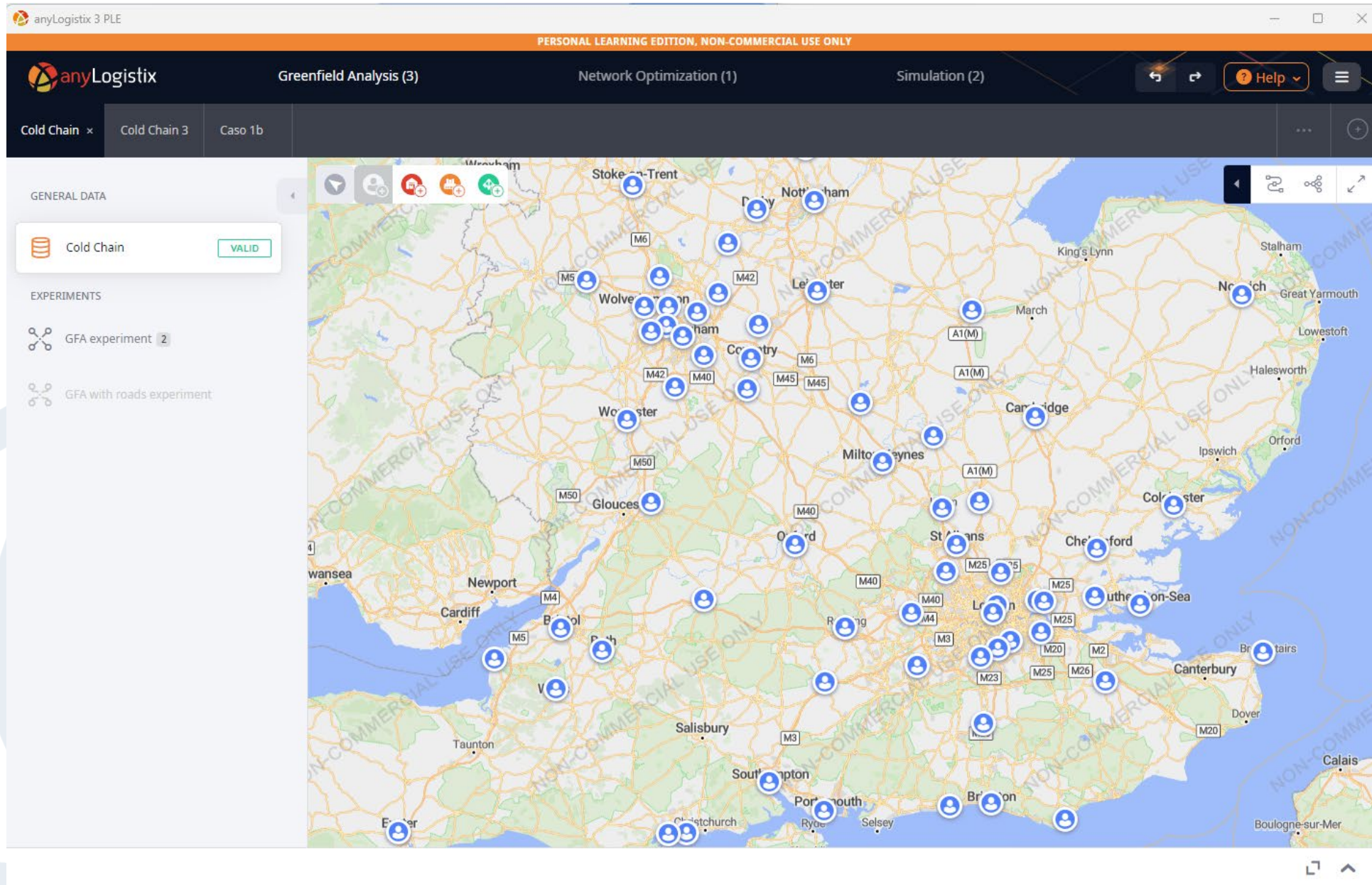
The screenshot shows a detailed view of the 'Customers' entity list. On the left, a sidebar lists available entities: 'Customers (100)', 'DCs and Factories', 'Demand (900)', 'Groups (1)', 'Location Groups', 'Locations (100)', and 'Period Groups'. The main area displays a table with columns: '#', 'Name', 'Type', 'Location', 'Inclusion Type', 'Additional Parameters', and 'Icon'. The table lists 7 customer locations: London, Birmingham, Liverpool, Nottingham, Sheffield, Bristol, and Leicester. Each row includes a dropdown menu for the location and a button for 'Additional parameters'. At the bottom, there are buttons for 'Add', 'Remove', and 'Generate'.

#	Name	Type	Location	Inclusion Type	Additional Parameters	Icon
1	London	Customer	London location	Include	Additional parameters	
2	Birmingham	Customer	Birmingham location	Include	Additional parameters	
3	Liverpool	Customer	Liverpool location	Include	Additional parameters	
4	Nottingham	Customer	Nottingham location	Include	Additional parameters	
5	Sheffield	Customer	Sheffield location	Include	Additional parameters	
6	Bristol	Customer	Bristol location	Include	Additional parameters	
7	Leicester	Customer	Leicester location	Include	Additional parameters	

List of elements of the selected entity: filtering, sorting, adding, deleting, etc.

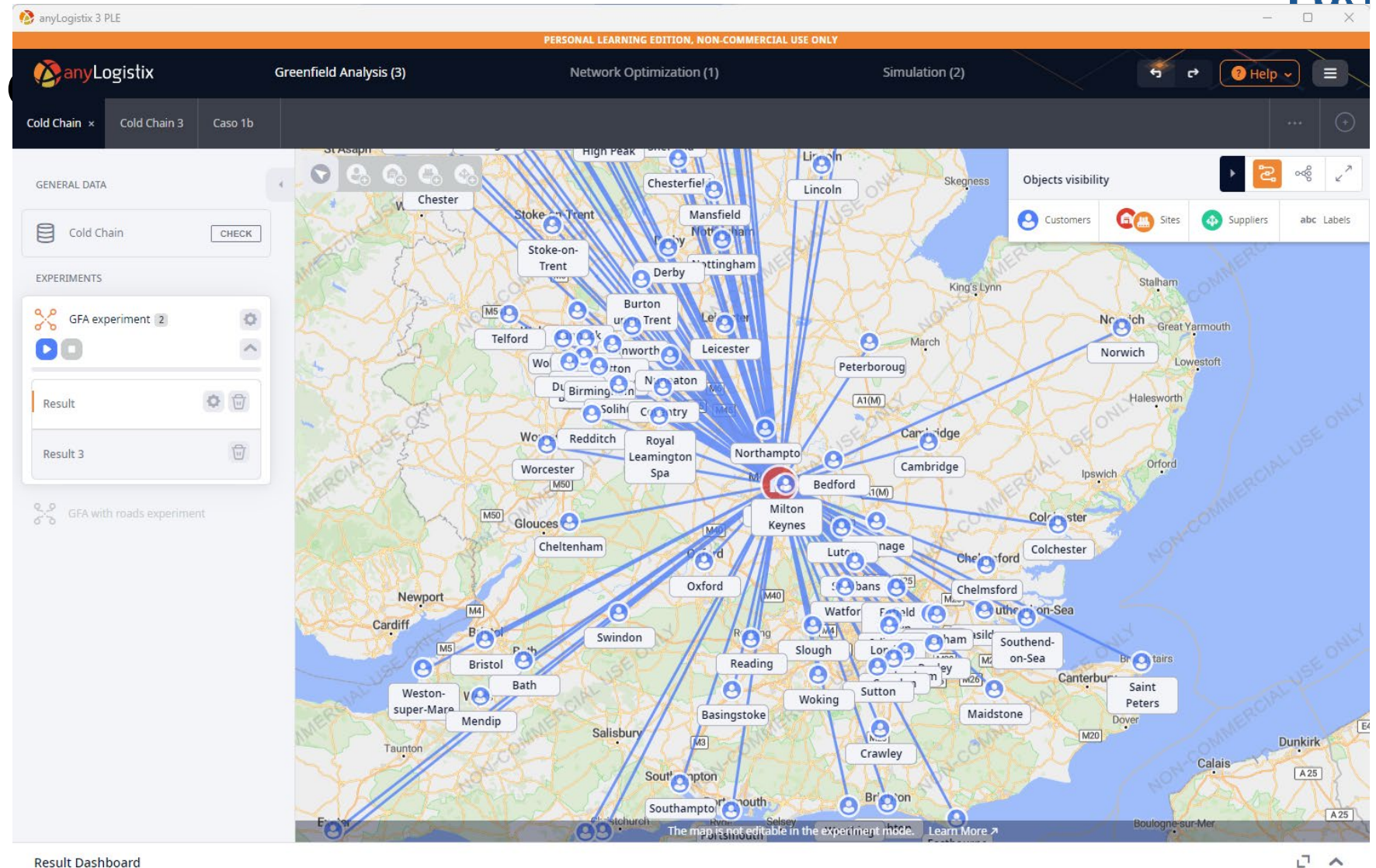


Enlarged geographic view



Extended geographic view + connections

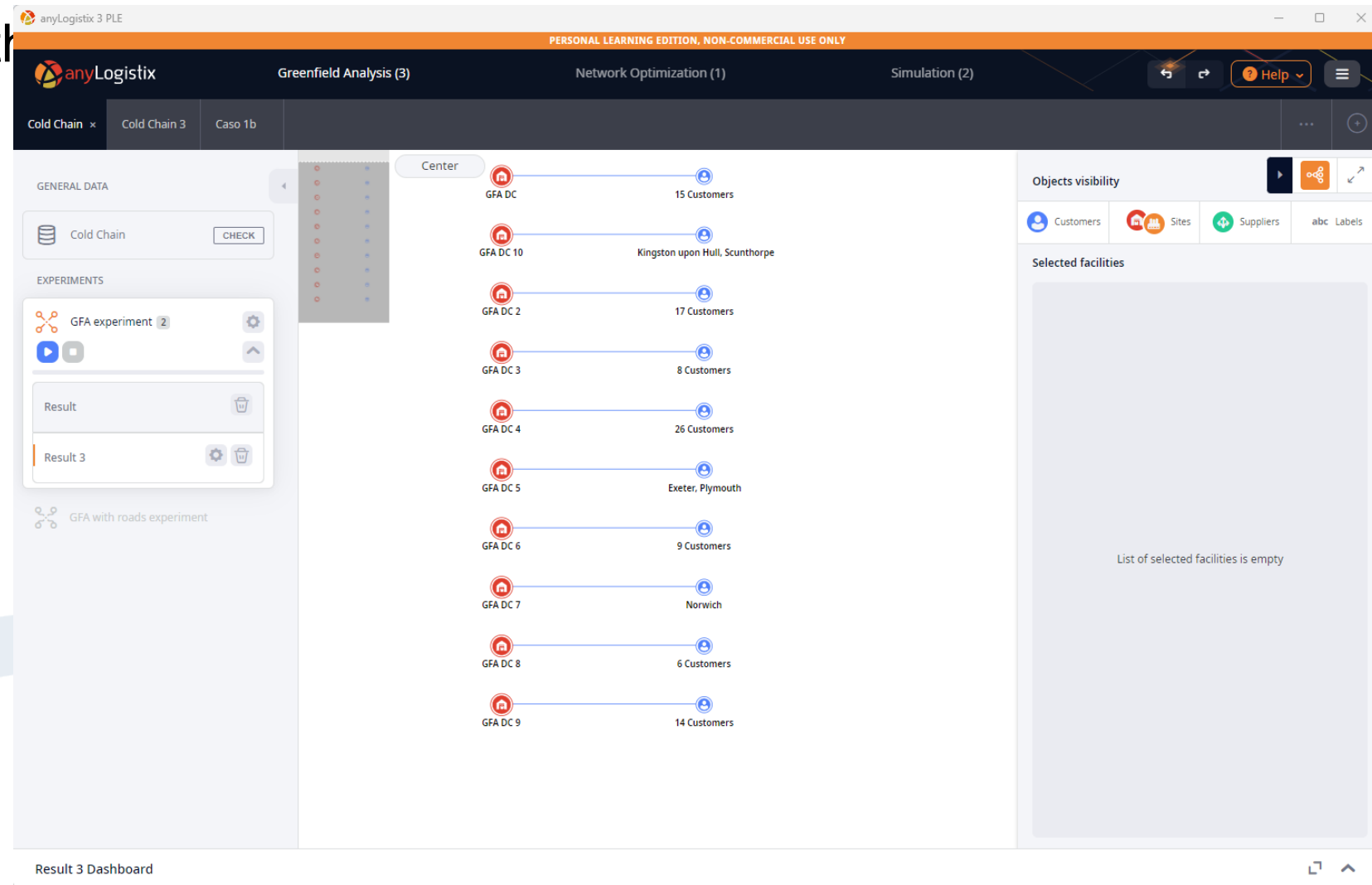
- After the





Logical view (with connections)

- After the





Version 2 allowed exporting the database.

anyLogistix PLE - Non-commercial use only - Caso 1

File Extensions Settings Help Feature Request ALX Community

GFA [1] NO **SIM [6]** TO

ALX2-Caso 2 GFA Result
ALX2-Caso 2 GFA Con proveedores por product
ALX2-Caso 2 con Baleares - Bueno
ALX2-Caso 2 con Canarias
ALX2-Caso 2 con Canarias - Bueno
ALX2-Caso 2 con Canarias - Mixto Baleares y C

+ New Scenario
Import Scenario from Excel
Import Scenario from Database

Basic All In use [102]

Customers [20]

DCs and Factories [8]
Demand [100]
Inventory [27]
Paths [14]
Periods [1]
Products [5]
Shipping [12]

Data

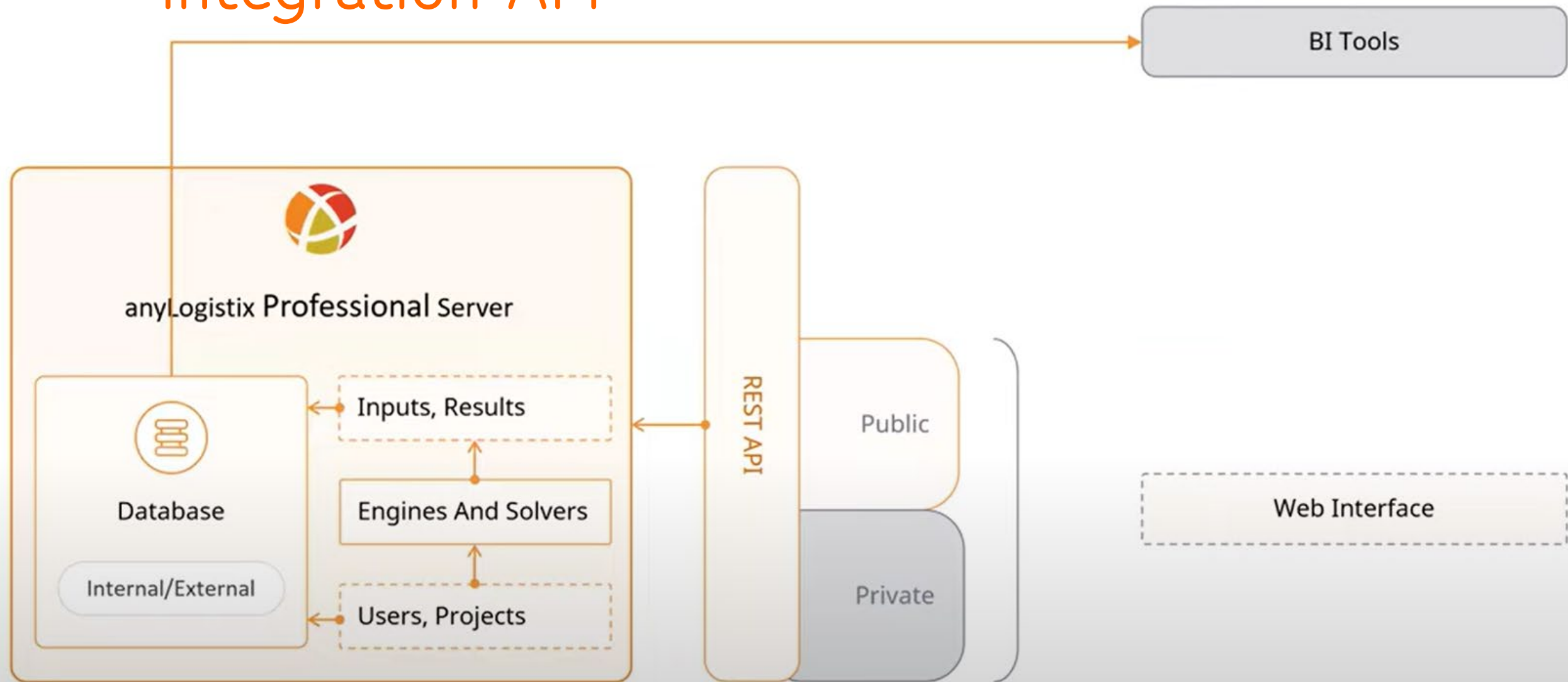
Simulation experiment
Variation experiment
Comparison experiment
Safety stock estimation
Risk analysis experiment
Custom experiment
External tables

Map showing a route through Northern Spain, including cities like Santiago de Compostela, Ourense, Pontevedra, and Valladolid.

#	Name	Type	Location	Inclusion Type	Additional Param...	Icon
1	Cli.Madrid	Customer	Cli.Madrid locati..	Include	Additional parame...	ⓘ
2	Cli.Barcelona	Customer	Cli.Barcelona loc..	Include	Additional parame...	ⓘ
3	Cli.Valencia	Customer	Cli.Valencia locat..	Include	Additional parame...	ⓘ
4	Cli.Sevilla	Customer	Cli.Sevilla locatior	Include	Additional parame...	ⓘ
5	Cli.Zaragoza	Customer	Cli.Zaragoza loca..	Include	Additional parame...	ⓘ
6	Cli.Málaga	Customer	Cli.Málaga locati..	Include	Additional parame...	ⓘ



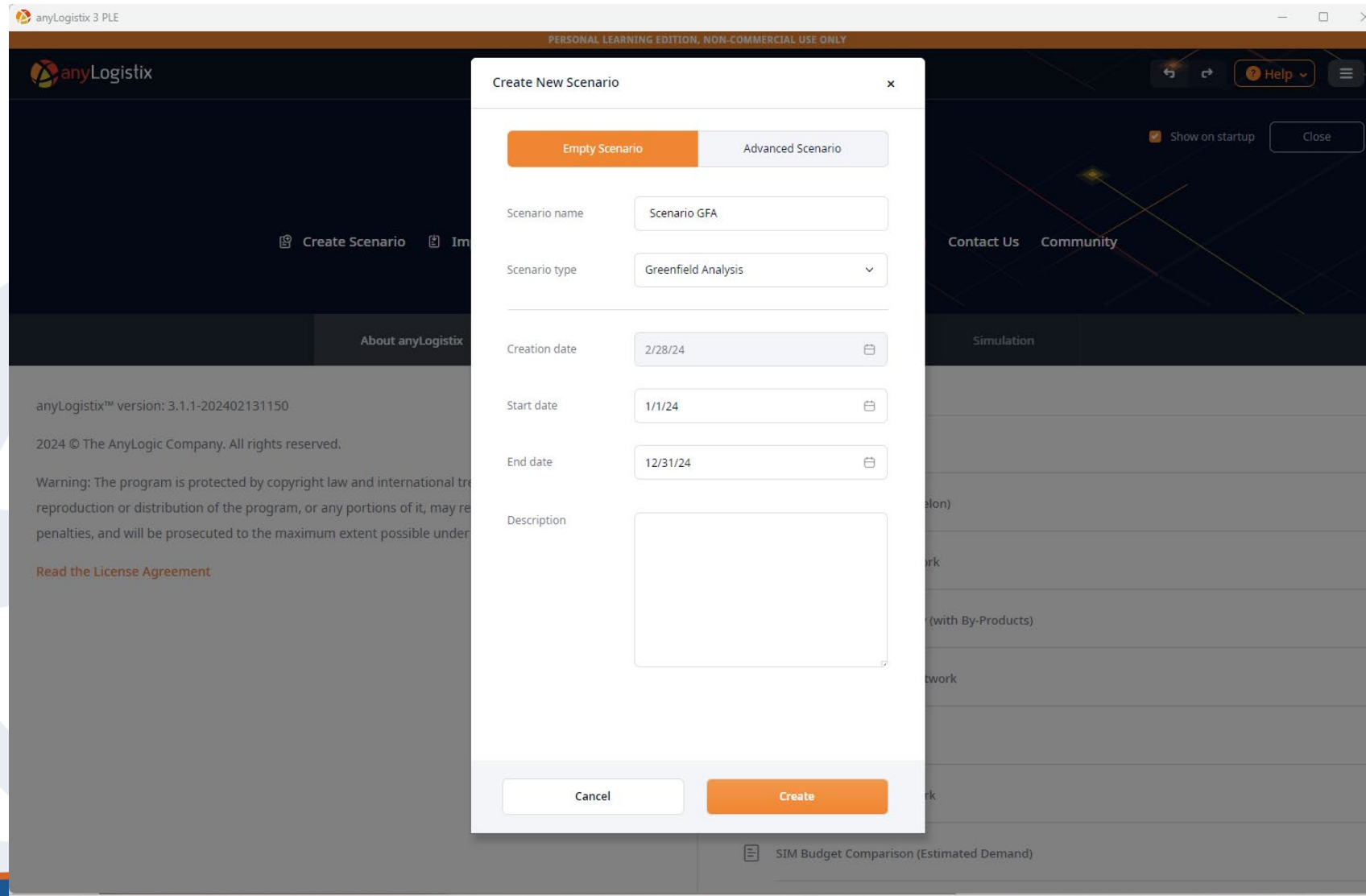
The PRO version incorporates the integration API





1. Creating a scenario...

- Simple option...



The screenshot displays the 'anyLogistix 3 PLE' software window. A 'Create New Scenario' dialog box is open in the center. The dialog has two tabs: 'Empty Scenario' (selected) and 'Advanced Scenario'. The 'Empty Scenario' tab contains the following fields:

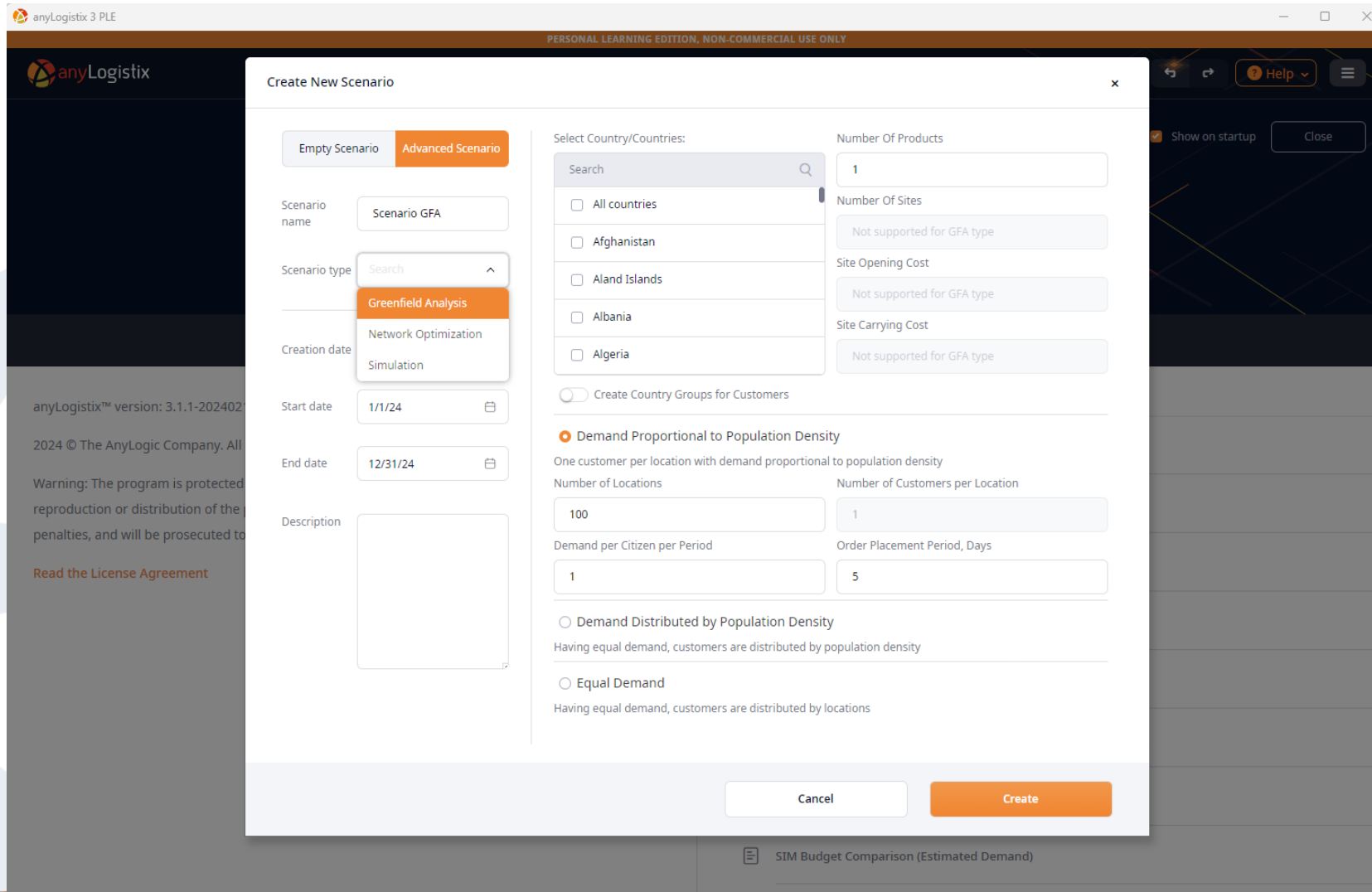
- Scenario name:** A text input field containing 'Scenario GFA'.
- Scenario type:** A dropdown menu set to 'Greenfield Analysis'.
- Creation date:** A date picker set to '2/28/24'.
- Start date:** A date picker set to '1/1/24'.
- End date:** A date picker set to '12/31/24'.
- Description:** A large text area for additional details.

At the bottom of the dialog are 'Cancel' and 'Create' buttons. The background software interface shows a dark theme with a sidebar on the left containing 'Create Scenario' and 'Im' (likely 'Import') buttons. The main area displays a simulation map with a yellow star and various labels like 'Contact Us', 'Community', and 'Simulation'. A status bar at the bottom indicates 'SIM Budget Comparison (Estimated Demand)'.



1. Creating a scenario...

- Advanced option...



anyLogistix 3 PLE

PERSONAL LEARNING EDITION, NON-COMMERCIAL USE ONLY

anyLogistix

Create New Scenario

Empty Scenario **Advanced Scenario**

Scenario name: Scenario GFA

Scenario type: Search

- Greenfield Analysis
- Network Optimization
- Simulation

Creation date: 1/1/24

Start date: 1/1/24

End date: 12/31/24

Description:

Select Country/Countries:

Search

- ☐ All countries
- ☐ Afghanistan
- ☐ Aland Islands
- ☐ Albania
- ☐ Algeria

☐ Create Country Groups for Customers

☒ Demand Proportional to Population Density

One customer per location with demand proportional to population density

Number of Locations: 100

Number of Customers per Location: 1

Demand per Citizen per Period: 1

Order Placement Period, Days: 5

☐ Demand Distributed by Population Density

Having equal demand, customers are distributed by population density

☐ Equal Demand

Having equal demand, customers are distributed by locations

Cancel Create

anyLogistix™ version: 3.1.1-202402

2024 © The AnyLogic Company. All

Warning: The program is protected reproduction or distribution of the penalties, and will be prosecuted to

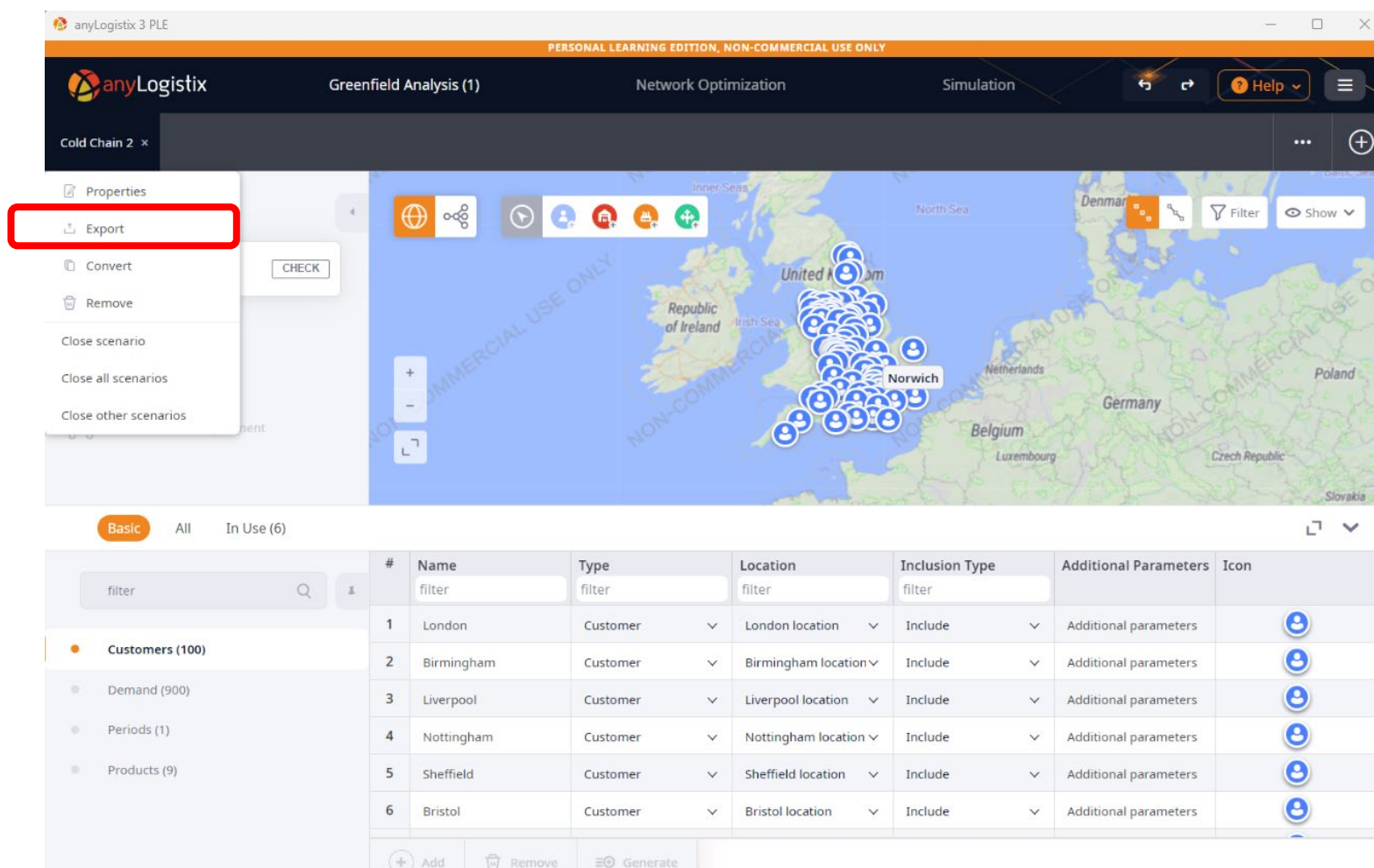
[Read the License Agreement](#)

SIM Budget Comparison (Estimated Demand)



2. Exporting the scenario

- By default, the scenarios are stored in the internal ALX database where we cannot edit them directly (it is possible but with a lot of "alchemy").
- We can export our scenario to an EXCEL file that is much easier to handle.



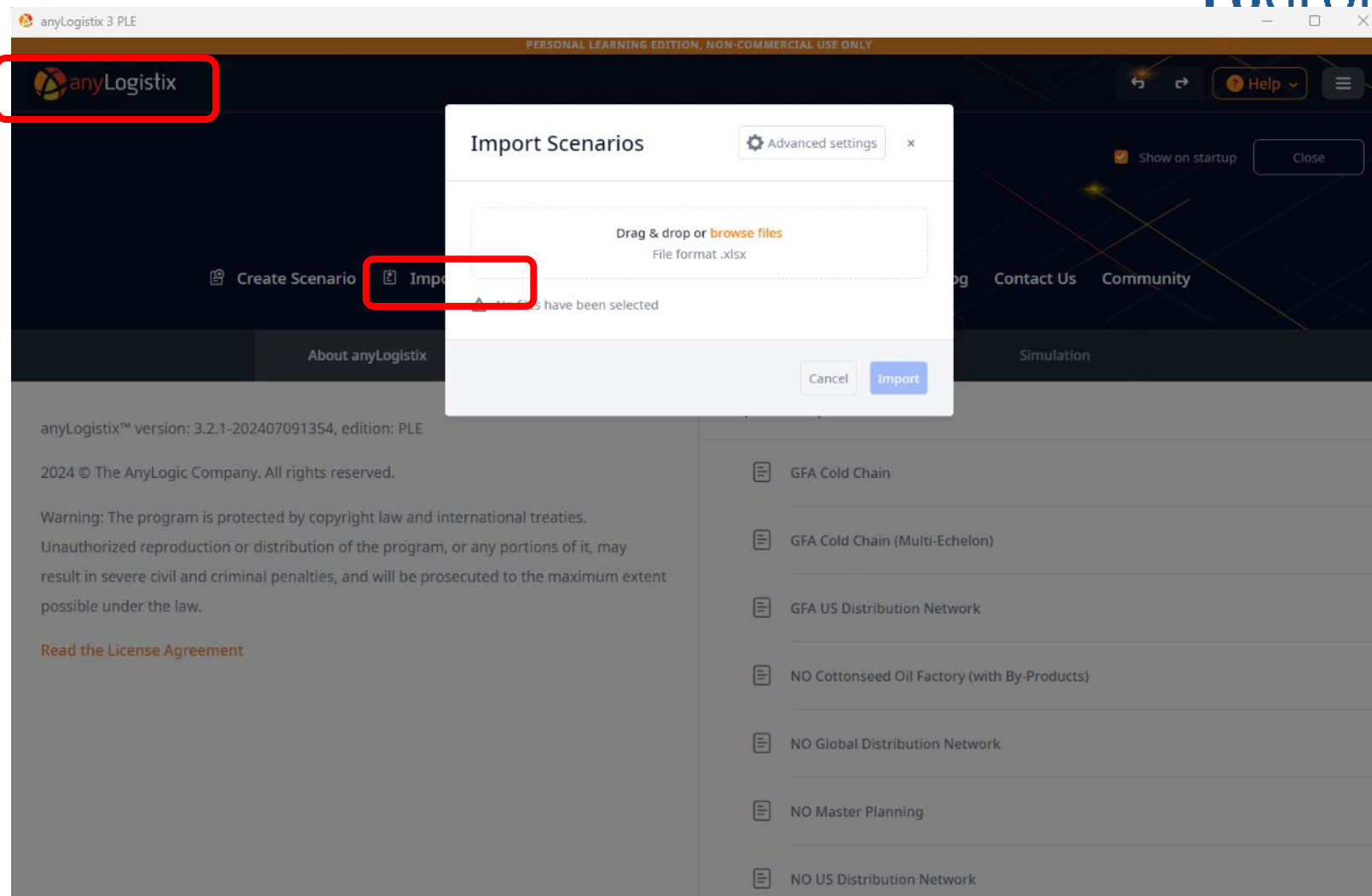
The screenshot displays the anyLogistix 3 PLE interface. A context menu is open over the map, with the 'Export' option highlighted by a red rectangle. The map shows a network of locations in the UK, including London, Birmingham, Liverpool, Nottingham, Sheffield, and Bristol. Below the map, a table lists the customer data for the scenario.

#	Name	Type	Location	Inclusion Type	Additional Parameters	Icon
1	London	Customer	London location	Include	Additional parameters	
2	Birmingham	Customer	Birmingham location	Include	Additional parameters	
3	Liverpool	Customer	Liverpool location	Include	Additional parameters	
4	Nottingham	Customer	Nottingham location	Include	Additional parameters	
5	Sheffield	Customer	Sheffield location	Include	Additional parameters	
6	Bristol	Customer	Bristol location	Include	Additional parameters	



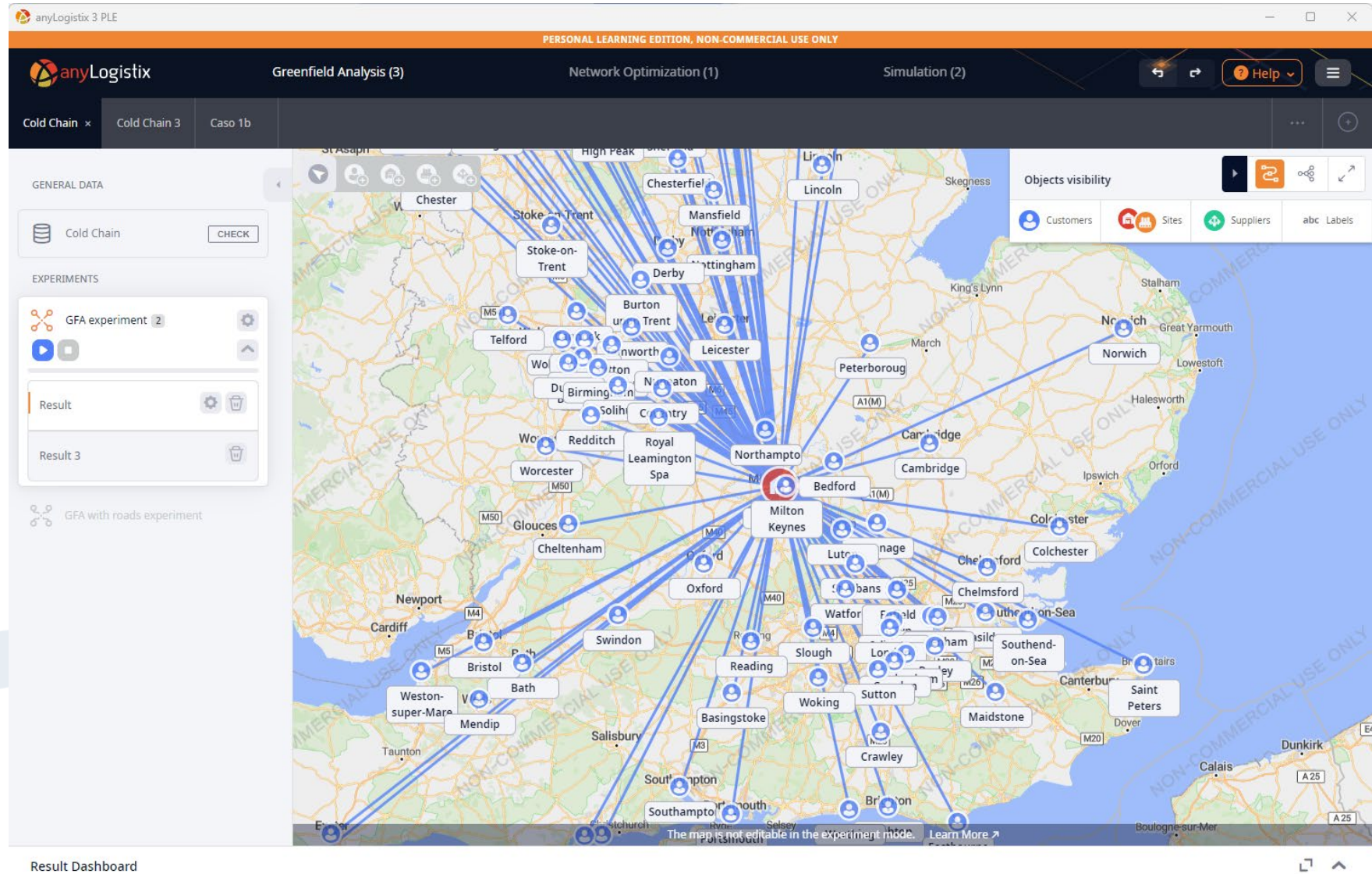
3. We import other scenarios

- Saved scenarios can later be imported from the main menu.





We developed the scenario in class...



Result Dashboard